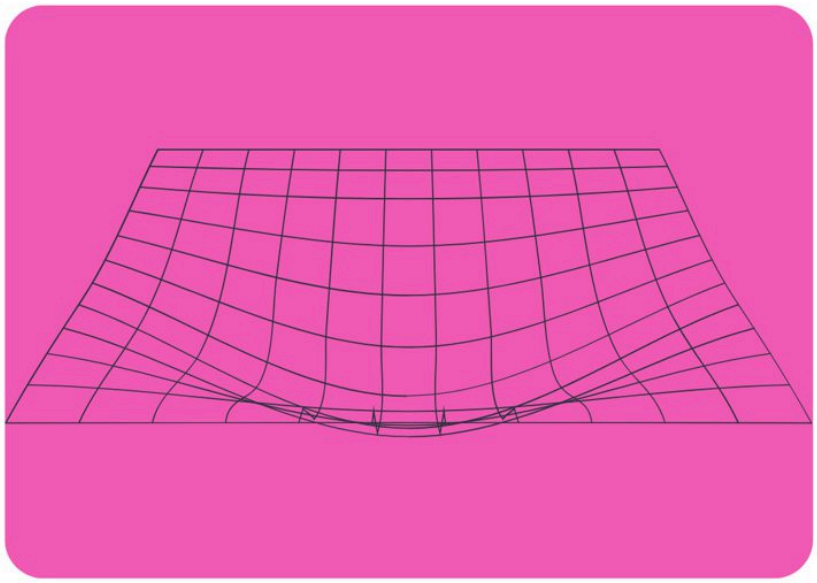
Chapter 17

Gravitational Fields



© James Gan

A Level Physics
Prepared and taught by James Gan

Chapter Outline

17.1 Newton's law of gravitation

17.2 Gravitational Field Strength (g)

17.3 Gravitational Potential

17.4 Motion in Gravitational Orbits

17.5 Calculating Orbital Periods

17.6 Earth's Orbital Mechanics



17.1 Newton's law of gravitation

Newton's law of gravitation

Definition

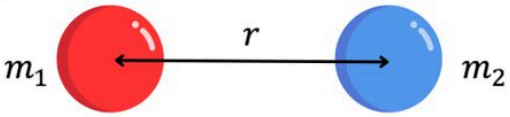
Every object with mass **attracts** every other mass with a **force** that is:

- Directly proportional to the product of their masses
- Inversely proportional to the square of the distance between them

Formula

$$F = G\frac{m_1 m_2}{r^2}$$

Diagram



Gravitational Constant

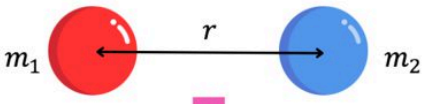
$6.674 \times 10^{-11}\ Nm^2/kg^2$

Why G is needed: It sets the scale of the gravitational force, converting the proportional relationship into an exact, measurable value in Newtons. Without G , we would only know how masses and distance affect the force but not the actual numerical force.

17.1 Newton's law of gravitation

Newton's law of gravitation

When applying Newton's law of gravitation, the distance ***r*** is measured from the center of mass of one object to the center of mass of the other.



Since we are calculating the distance from an object's center of mass, we can essentially treat each object as a **point-like mass**.



17.1 Newton's law of gravitation



Newton's Third Law

The gravitational force that object 1 exerts on object 2 is equal in magnitude and opposite in direction to the force that object 2 exerts on object 1.

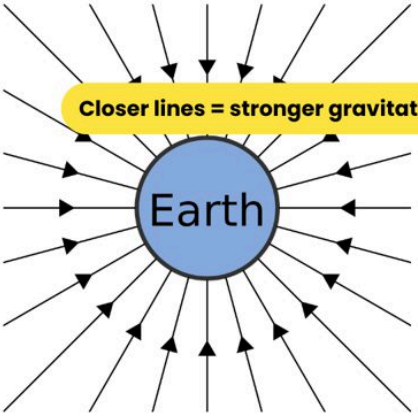
Both objects pull on each other with the same force, regardless of their size.

However, because ***Force = Mass × Acceleration***, the apple moves more noticeably than the Earth because of its much smaller mass. The Earth, with its enormous mass, experiences an extremely small acceleration, making its movement negligible.

17.1 Newton's law of gravitation

Visualizing Gravitational Fields

- Gravitational fields are represented by **field lines**.
- Arrows on the lines indicate the **direction of the gravitational force (towards the mass)**.



Closer lines = stronger gravitational field.

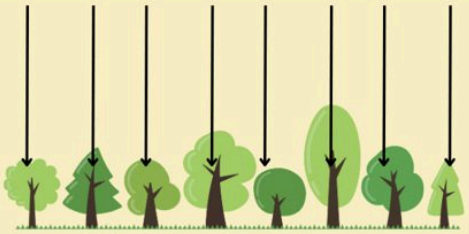
Wider spaced lines = Weaker gravitational field.

17.1 Newton's law of gravitation

Uniform vs. Radial Gravitational Fields

Near Earth's Surface

- The gravitational field is uniform.
- Field lines are parallel and equally spaced.
- Gravity pulls straight down with the same strength everywhere.



Far From Earth

- The gravitational field is radial.
- Field lines spread out from Earth's center.
- Gravity points toward Earth's center and weakens with distance.



17.1 Newton's law of gravitation

The Inverse Square Law

- Gravitational force decreases with the square of the distance between objects.
- If the distance ***r*** is doubled, the force becomes 4 times weaker:

$$F \propto \frac{1}{r^2}$$



This explains why gravity is stronger near planets and weaker in space.

17.2 Gravitational Field Strength (g)

Gravitational Field Strength

Definition

Gravitational field strength ***(g)*** is the **force per unit mass** experienced by an object in a gravitational field.

Unit

m/s^2 , N/kg

In Relation to Weight

Gravitational field strength is directly related to the weight of an object. **The weight is the force exerted by gravity on an object**:

$$w = mg$$

17.2 Gravitational Field Strength (g)

Equation for Gravitational Field Strength

$$w = mg$$

$$F = mg$$

$$g = \frac{F}{m}$$

$$F = G\frac{m_1 m_2}{r^2}$$

$$g = \frac{G\frac{mM}{r^2}}{m} = \frac{GmM}{r^2 m} = \frac{GM}{r^2}$$

© James Gan

17.2 Gravitational Field Strength (g)

Equation for Gravitational Field Strength

$$w = mg$$

$$F = mg$$

$$g = \frac{F}{m}$$

$$F = G\frac{m_1 m_2}{r^2}$$

$$g = \frac{G\frac{mM}{r^2}}{m} = \frac{GmM}{r^2 m} = \frac{GM}{r^2}$$

Final Formula

$$g = \frac{GM}{r^2}$$

17.2 Gravitational Field Strength (g)

Equation for Gravitational Field Strength

$$w = mg$$

$$F = mg$$

$$g = \frac{F}{m}$$

$$F = G\frac{m_1 m_2}{r^2}$$

$$g = \frac{G\frac{mM}{r^2}}{m} = \frac{GmM}{r^2 m} = \frac{GM}{r^2}$$

Final Formula

$$g = \frac{GM}{r^2}$$

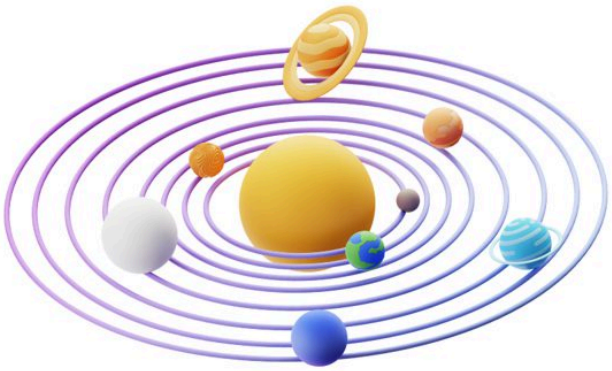
Why is this derivation important?

- By deriving this equation, we can calculate the **gravitational field strength** at any point around a mass, such as Earth.
- This helps predict how objects will move in a gravitational field, which is crucial for satellite orbits, object motion, and space missions.
- The mass ***m*** of an object placed in the field does not appear in the equation for gravitational field strength because ***g* is the strength of the field itself, independent of the object's mass.**

17.2 Gravitational Field Strength (g)

Gravitational Field Strength as a Vector Quantity

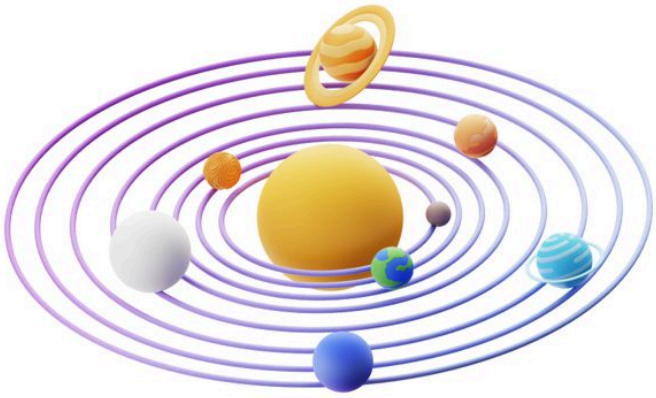
- Gravitational field strength is a **vector quantity**, meaning it has both magnitude and direction.
- The direction of the gravitational field strength is always toward the **center** of the mass creating the field.



17.2 Gravitational Field Strength (g)

Inverse square law

- The field strength **decreases** as the distance **increases** from the source of gravity.



- This means that as you move farther from the source mass, the gravitational field strength weakens.
- This follows from the equation:

$$g = \frac{GM}{r^2}$$

17.2 Gravitational Field Strength (g)

Acceleration Due to Gravity (Free Fall)

Acceleration of free fall is the acceleration experienced by an object in a gravitational field when no other forces (like air resistance) are acting on it.



It is equal to the gravitational field strength at that location.

Near Earth's surface, the acceleration due to gravity is approximately:

$$g = 9.8\ m/s^2$$

17.2 Gravitational Field Strength (g)

Worked Example

What is the gravitational field strength 10 km above Earth's surface?
Given that:

$Mass\ of\ Earth = 5.98 \times 10^{24} kg$

$Radius\ of\ Earth = 6370\ km$

$Gravitational\ Constant = 6.67\ x\ 10^{-11}\ Nm^2kg^{-2}$



Solution

$$g = \frac{GM}{r^2}$$

17.2 Gravitational Field Strength (g)

Worked Example

What is the gravitational field strength 10 km above Earth's surface?
Given that:

$Mass\ of\ Earth = 5.98 \times 10^{24} kg$

$Radius\ of\ Earth = 6370\ km$

$Gravitational\ Constant = 6.67\ x\ 10^{-11}\ Nm^2kg^{-2}$



Solution

$$g = \frac{GM}{r^2}$$

$$g = \frac{(6.67\ x\ 10^{-11}) \times (5.98 \times 10^{24})}{(6370 + 10)^2} = 9.81\ N/kg$$

17.2 Gravitational Field Strength (g)

Worked Example

What is the gravitational field strength 500 km above Earth's surface?
Given that:

$Mass\ of\ Earth = 5.98 \times 10^{24} kg$

$Radius\ of\ Earth = 6370\ km$

$Gravitational\ Constant = 6.67\ x\ 10^{-11}\ Nm^2kg^{-2}$



Solution

$$g = \frac{GM}{r^2}$$

17.2 Gravitational Field Strength (g)

Worked Example

What is the gravitational field strength 500 km above Earth's surface?
Given that:

$Mass\ of\ Earth = 5.98 \times 10^{24} kg$

$Radius\ of\ Earth = 6370\ km$

$Gravitational\ Constant = 6.67\ x\ 10^{-11}\ Nm^2kg^{-2}$



Solution

$$g = \frac{GM}{r^2}$$

$$g = \frac{(6.67\ x\ 10^{-11}) \times (5.98\ \times\ 10^{24})}{(6370 + 500)^2} = 8.44\ N/kg$$

17.3 Gravitational Potential

Limitations of *mgh* for Gravitational Potential Energy

- The equation PE = *mgh* is useful ***only*** near Earth's surface, where g is constant.
- However, as an object moves further from Earth, **g decreases** because of the inverse square law.
- This means that using **h = 0 at Earth's surface is an oversimplification**, as potential energy should be measured relative to a more fundamental reference point.
- To properly account for the change in **g** with distance, we need a more general formula for gravitational potential energy.

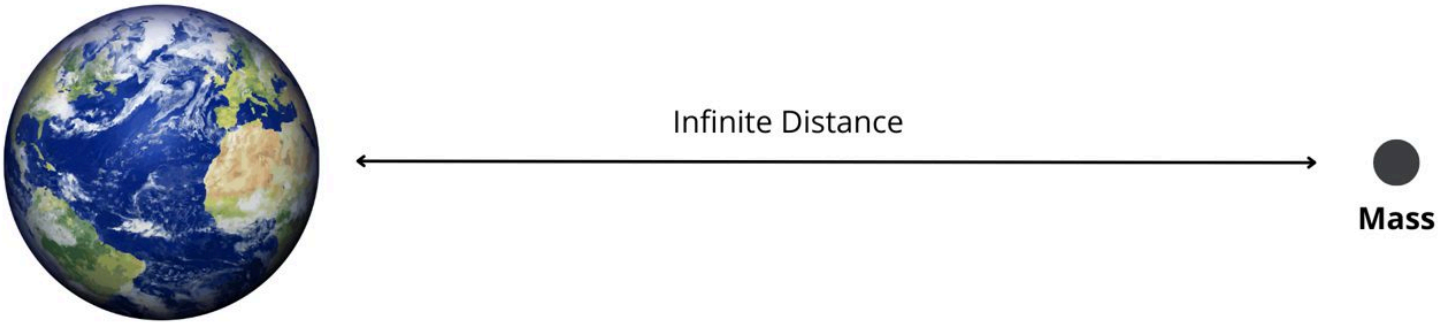


© James Gan

17.3 Gravitational Potential

Defining Gravitational Potential Energy at Infinity

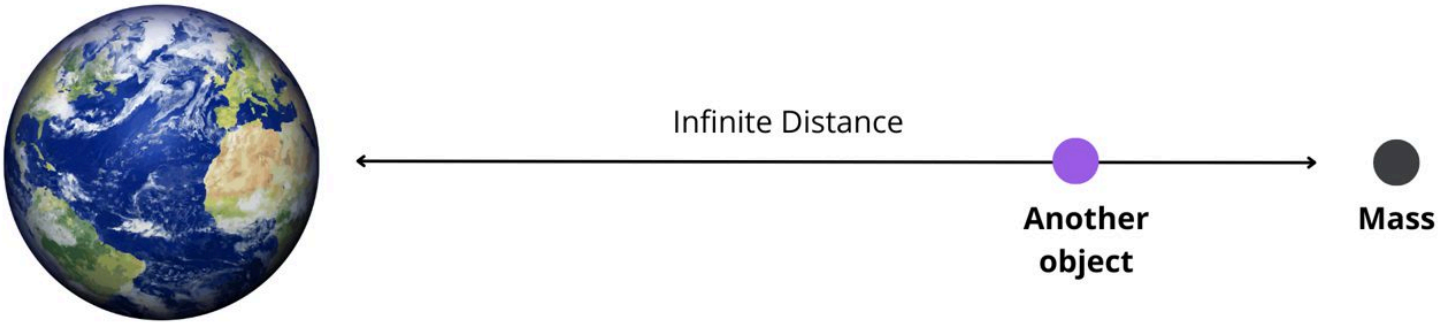
- Imagine a mass very, very far away from everything—so far that gravity has no effect on it.
- We say that at this infinite distance, **the mass has zero potential energy.**



17.3 Gravitational Potential

Defining Gravitational Potential Energy at Infinity

- This makes calculations easier and more universal because any object closer than infinity will have **negative potential energy (since gravity is pulling it back)**.



17.3 Gravitational Potential

Gravitational Potential

Definition

Gravitational potential (Φ) at a point is the work done per unit mass to bring a small test mass from infinity to that point.

Formula

$$\Phi = -\frac{GM}{r}$$

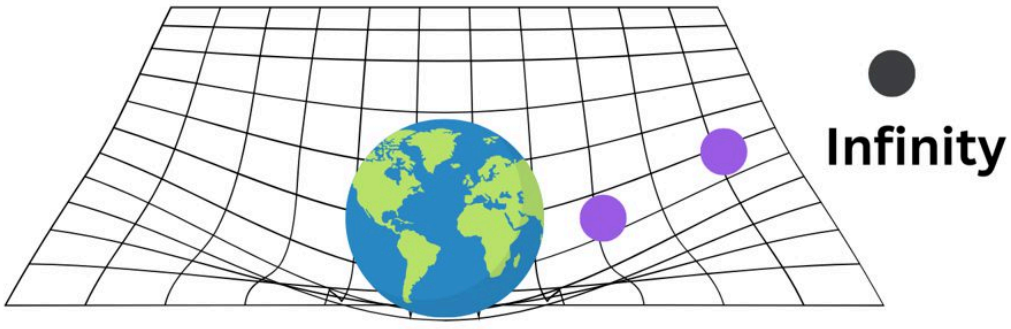
Key Idea

Describes how much potential energy each kg of mass would have at a certain point in a gravitational field.

17.3 Gravitational Potential

Formula

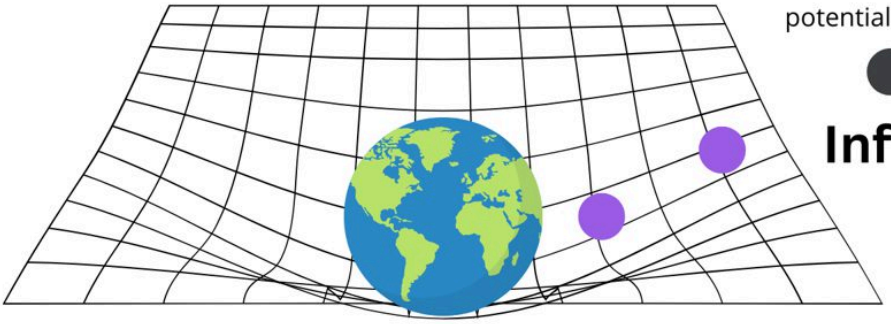
$$\Phi = -\frac{GM}{r}$$



17.3 Gravitational Potential

Why is GP Always Negative?

Zero at infinity: By definition, gravitational potential is zero at infinity.



Infinity

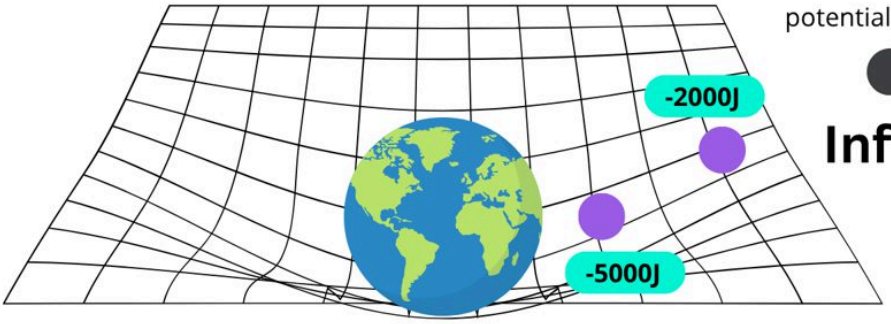
Idea: Objects in a gravitational field are "trapped":

- To move a mass from infinity to a point near a planet, gravity does work on it, pulling it inward.
- This means the object loses potential energy, making the value negative.

17.3 Gravitational Potential

Why is GP Always Negative?

Zero at infinity: By definition, gravitational potential is zero at infinity.



Infinity

Closer = More Negative: The closer an object is to the planet, the more energy is needed to escape, making the stored energy more negative.

More negative = deeper energy well = more energy required to escape.

17.3 Gravitational Potential

Worked Example: Comparing Gravitational Potential of Earth and Sun

Compare the gravitational potential at:

1. The surface of the Earth
2. The surface of the Sun

Info:

$$r_E = 6.37 \times 10^6\ m$$

$$M_E = 5.97 \times 10^{24}\ kg$$

$$r_S = 6.96 \times 10^8\ m$$

$$M_S = 1.99 \times 10^{30}\ kg$$

$$G = 6.674 \times 10^{-11}\ Nm^2/kg^2$$

Answer:

17.3 Gravitational Potential

Worked Example: Comparing Gravitational Potential of Earth and Sun

Compare the gravitational potential at:

1. The surface of the Earth
2. The surface of the Sun

Info:

$$r_E = 6.37 \times 10^6\ m$$

$$M_E = 5.97 \times 10^{24}\ kg$$

$$r_S = 6.96 \times 10^8\ m$$

$$M_S = 1.99 \times 10^{30}\ kg$$

$$G = 6.674 \times 10^{-11}\ Nm^2/kg^2$$

Answer:

Earth:

$$\Phi = -\frac{GM}{r}$$

$$= -\frac{(6.674 \times 10^{-11}) \times (5.97 \times 10^{24})}{6.37 \times 10^6}$$

$$= -6.26 \times 10^7\ J/kg$$



17.3 Gravitational Potential

Worked Example: Comparing Gravitational Potential of Earth and Sun

Compare the gravitational potential at:

1. The surface of the Earth
2. The surface of the Sun

Info:

$$r_E = 6.37 \times 10^6\ m$$

$$M_E = 5.97 \times 10^{24}\ kg$$

$$r_S = 6.96 \times 10^8\ m$$

$$M_S = 1.99 \times 10^{30}\ kg$$

$$G = 6.674 \times 10^{-11}\ Nm^2/kg^2$$

Answer:

Earth:

$$\Phi = -\frac{GM}{r}$$

$$= -\frac{(6.674 \times 10^{-11}) \times (5.97 \times 10^{24})}{6.37 \times 10^6}$$

$$= -6.26 \times 10^7\ J/kg$$

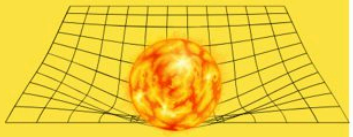


Sun:

$$\Phi = -\frac{GM}{r}$$

$$= -\frac{(6.674 \times 10^{-11}) \times (1.99 \times 10^{30})}{6.96 \times 10^8}$$

$$= -1.91 \times 10^9\ J/kg$$



17.3 Gravitational Potential

Worked Example: Comparing Gravitational Potential of Earth and Sun

A **500 kg satellite** is launched from the Earth's surface into a circular orbit at an altitude of **500 km** above the Earth. Calculate the energy required to move the satellite from the Earth's surface to its orbital height.

Info:

$$r_E = 6.37 \times 10^6\ m$$

$$M_E = 5.97 \times 10^{24}\ kg$$

$$G = 6.674 \times 10^{-11}\ Nm^2/kg^2$$

Answer:

© James Gan

17.3 Gravitational Potential

Worked Example: Comparing Gravitational Potential of Earth and Sun

A **500 kg satellite** is launched from the Earth's surface into a circular orbit at an altitude of **500 km** above the Earth. Calculate the energy required to move the satellite from the Earth's surface to its orbital height.

Info:

$r_E = 6.37 \times 10^6\ m$

$M_E = 5.97 \times 10^{24}\ kg$

$G = 6.674 \times 10^{-11}\ Nm^2/kg^2$

Answer:

$$\Delta \emptyset = \frac{GM}{r_1} - \frac{GM}{r_2}$$

$$\Delta \emptyset = GM\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

$$= 6.674 \times 10^{-11} \times 5.97 \times 10^{24} \times \left(\frac{1}{6.37 \times 10^6} - \frac{1}{6.87 \times 10^6}\right)$$

$$= 2.19 \times 10^8\ J$$

17.3 Gravitational Potential

Comparing Gravitational Field Strength and Gravitational Potential

	Gravitational Field Strength	Gravitational Potential
Definition	Defined as the force per unit mass experienced by a small test mass in the field.	Defined as the work done per unit mass to bring a small test mass from infinity to a point in the field.
Formula	$g = \frac{GM}{r^2}$	$\Phi = -\frac{GM}{r}$
Types	Vector Quantity	Scalar Quantity
Units	N/kg	J/kg

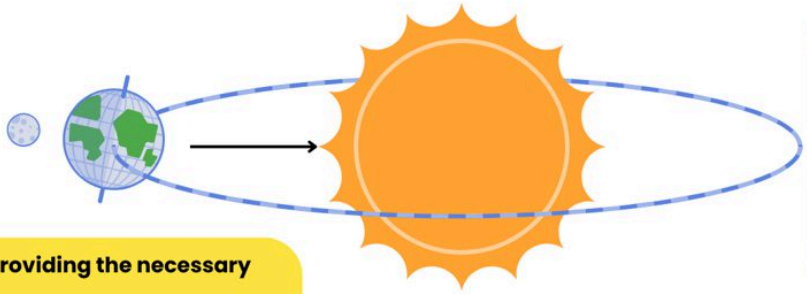
© James Gan

17.4 Motion in Gravitational Orbits

Deriving $v^2 = \frac{GM}{r}$

Step 1: Understanding the Forces Acting on an Orbiting Object

- An object in a circular orbit experiences:



Centripetal Force (needed to keep the object moving in a circle):

$$F = \frac{mv^2}{r}$$

Gravitational Force (providing the necessary centripetal force):

$$F = G\frac{m_1 m_2}{r^2}$$

17.4 Motion in Gravitational Orbits

Deriving $v^2 = \frac{GM}{r}$

Step 2: Equating Gravitational Force to Centripetal Force

- Since gravity is the only force keeping the object in orbit, we set the two equations equal, and simplify:

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

cancel out the m

$$\frac{GM}{r^2} = \frac{v^2}{r}$$

multiply by r

$$v^2 = \frac{GM}{r}$$

This means that all satellites at the same orbital radius around Earth will have the same speed, regardless of their mass, ensuring predictable and stable orbits for objects like the ISS, GPS satellites, and space debris.

17.4 Motion in Gravitational Orbits

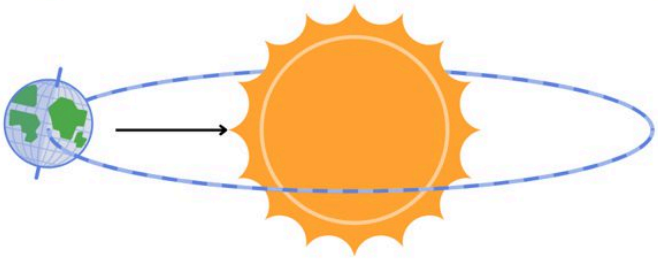
Worked Example

Calculate the orbital speed of the Earth around the Sun.

mass of the sun: $M = 2 \times 10^{30}\ kg$

Average distance between Earth and Sun: $r = 1.5 \times 10^{11}\ m$

$G = 6.674 \times 10^{-11}\ Nm^2/kg^2$



$$v^2 = \frac{GM}{r}$$

17.4 Motion in Gravitational Orbits

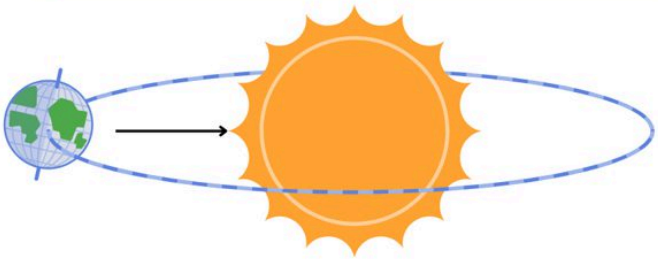
Worked Example

Calculate the orbital speed of the Earth around the Sun.

mass of the sun: $M = 2 \times 10^{30}\ kg$

Average distance between Earth and Sun: $r = 1.5 \times 10^{11}\ m$

$G = 6.674 \times 10^{-11}\ Nm^2/kg^2$



$$v^2 = \frac{GM}{r}$$

$$v = \sqrt{\frac{6.674 \times 10^{-11} \times 2 \times 10^{30}}{1.5 \times 10^{11}}}$$

$$= 2.97 \times 10^4 = 29.7\ km/s$$

17.5 Calculating Orbital Periods

Orbital Period

Definition

Orbital period is the time taken for a satellite to complete one full orbit around a mass.

Deriving the formula

$$s = \frac{d}{t}$$

$$v = \frac{2\pi r}{T}$$

$$v^2 = \frac{GM}{r}$$

$$\frac{4\pi^2 r^2}{T^2} = \frac{GM}{r}$$

$$T^2 = \frac{4\pi^2 r^3}{GM}$$

$$T = \sqrt{\frac{4\pi r^3}{GM}}$$

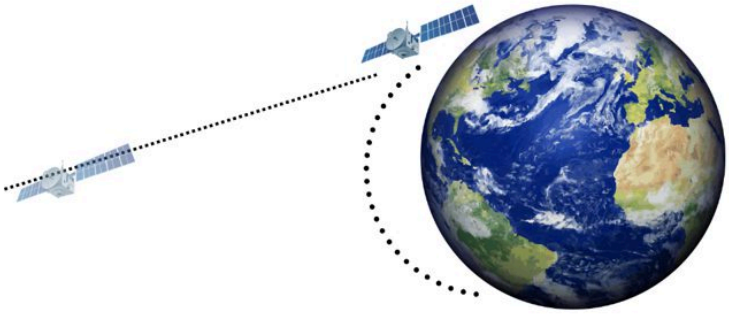
Importance

This formula is important because it allows us to predict how long an object will take to orbit a planet or star, helping in satellite deployment, space missions, and understanding planetary motion.

17.6 Earth's Orbital Mechanics

Orbital Speed and Radius

- For an object to maintain a stable orbit, it must travel at the correct speed for its altitude.
- From $v = \sqrt{\frac{GM}{r}}$, a smaller orbital radius requires a higher speed to counteract gravity.
- If an object moves too fast, it may escape orbit; if too slow, it may fall back to Earth.



17.6 Earth's Orbital Mechanics

Uses of Artificial Satellites



Communication: Enabling global connectivity, TV broadcasts, and internet services.



Navigation: GPS satellites provide location tracking.



Weather Monitoring: Satellites track storms, climate, and environmental changes.



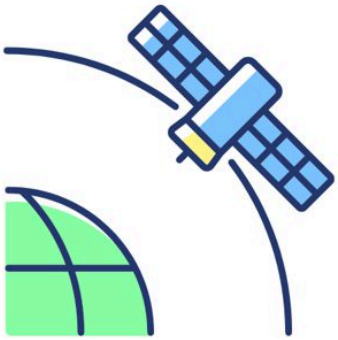
Military and Surveillance: Used for reconnaissance and national security.

17.6 Earth's Orbital Mechanics

Geostationary Orbits

A geostationary satellite orbits Earth once every 24 hours, staying fixed over a single location.

Usage: Ideal for telecommunications, weather observation, and broadcasting.



17.6 Earth's Orbital Mechanics

Determining Satellite Distance Using Orbital Period

$$T^2 = \frac{4\pi^2 r^3}{GM}$$

Step 1: Rearrange our equation so that r (distance) becomes the subject.

$$r = \sqrt[3]{\frac{GMT^2}{4\pi^2}}$$

© James Gan

17.6 Earth's Orbital Mechanics

Determining Satellite Distance Using Orbital Period

$$T^2 = \frac{4\pi^2 r^3}{GM}$$

Step 1: Rearrange our equation so that r (distance) becomes the subject.

$$r = \sqrt[3]{\frac{GMT^2}{4\pi^2}}$$

Step 2: A geostationary satellite has an orbital period of 24 hours or 86,400 seconds.

$$M_E = 5.97 \times 10^{24}\ kg$$
$$G = 6.674 \times 10^{-11}\ Nm^2/kg^2$$
$$T = 86400s$$

$$r = \sqrt[3]{\frac{GMT^2}{4\pi^2}}$$
$$= \sqrt[3]{\frac{(6.674 \times 10^{-11}) \times (5.97 \times 10^{24}) \times (86400)^2}{4\pi^2}}$$

Distance from the centre of the Earth ~ 42300km